SENTIMENT ANALYSIS OF INDONESIAN SPOTIFY REVIEWS USING MACHINE LEARNING AND BILSTM

Uliano Wilyam Purba
Department of Data Science
Faculty of Science
Sumatera Institute of Technology
Lampung, Indonesia
uliano.122450098@student.itera.ac.id

Andre Hadiman Rotua Parhusip
Department of Data Science
Faculty of Science
Sumatera Institute of Technology
andre.122450108@student.itera.ac.id

Sahid Maulana
Department of Data Science
Faculty of Science
Sumatera Institute of Technology
sahid.122450109@student.itera.ac.id

Luluk Muthoharoh M. Si
Department of Data Science
Faculty of Science
Sumatera Institute of Technology
luluk.muthoharoh@sd.itera.ac.id

Ardika Satria M. Si
Department of Data Science
Faculty of Science
Sumatera Institute of Technology
ardika.satria@sd.itera.ac.id

Martin C.T. Manullang Ph.D.
Department of Informatics Engineering
Faculty of Industrial Technology
Sumatra Institute of Technology
martin.manullang@if.itera.ac.id

May 2, 2026

ABSTRACT

This paper compares three Scikit-learn classifiers — Support Vector Machine (SVM), Multinomial Naive Bayes (MNB), and Decision Tree (DT) — with a two-layer Bidirectional Long Short-Term Memory (BiLSTM) model for three-class sentiment classification of Indonesian Spotify reviews. From 100,000 scraped reviews, 70,155 cleaned samples are used for the machine learning track, while the BiLSTM is trained on a stratified 20,000-sample subset because training is performed in a CPU-only environment. Both tracks use the same preprocessing pipeline: slang normalization, stopword removal, and Sastrawi stemming. In 5-fold cross-validation, Decision Tree is the strongest classical model with weighted F1-score 0.7269, ahead of SVM (0.6958) and MNB (0.4619). On its held-out test set, BiLSTM reaches weighted F1-score 0.8069, but fails on the minority Neutral class (F1 = 0.000). These results show that BiLSTM provides stronger overall performance, whereas ML with SMOTE yields more balanced three-class recognition.

Keywords Sentiment Analysis · Indonesian NLP · BILSTM · TF-IDF · Decision Tree · Google Play Store Reviews · PyCaret · SMOTE · Class Imbalance · Hugging Face Deployment

1 Introduction

Sentiment analysis is valuable for app developers because user reviews reveal satisfaction, complaints, and feature requests at scale. For Indonesian reviews, this task is harder because the text is highly informal, rich in slang, and often inconsistent in spelling.

Spotify reviews from the Google Play Store provide a relevant case study for comparing classical machine learning and deep learning on Indonesian text. In this work, SVM, Multinomial Naive Bayes, and Decision Tree with TF-IDF are benchmarked against a two-layer BiLSTM. The comparison is practical rather than perfectly symmetric: classical models use the full 70,155 cleaned samples, while BiLSTM is trained on a 20,000-sample stratified subset due to CPU-only constraints.

This study contributes: (1) a shared Indonesian preprocessing pipeline for both tracks; (2) a 5-fold benchmark of three classical classifiers with TF-IDF and SMOTE; (3) evaluation of a compact BiLSTM with 1,096,579 parameters; and (4) a transparent comparison of overall and per-class performance, supported by public deployment of both models.

- ML demo: `https://huggingface.co/spaces/tubespba-kelompoktuwir/sentimen-spotify-ml`
- DL demo: `https://huggingface.co/spaces/tubespba-kelompoktuwir/sentimen-spotify-dl`

2 Related Work

2.1 Sentiment Analysis: Foundational Work

Pang and Lee [1] established the main tasks and evaluation principles of sentiment analysis. Their survey also showed that classical supervised models with bag-of-words or TF-IDF features remain strong baselines for document-level polarity classification.

2.2 Classical Machine Learning for Text Classification

TF-IDF combined with SVM, Multinomial Naive Bayes, and tree-based models is widely used in sentiment classification. Prior studies on app reviews and social media show that SVM is often strong, while performance still depends heavily on feature design, label balance, and language characteristics [2, 3].

2.3 Deep Learning for Sentiment Analysis

LSTM and BiLSTM models are designed to capture sequential context more effectively than bag-of-words features. Prior work reports that BiLSTM performs well for sentiment tasks and can outperform classical methods on Indonesian text, although the gain usually comes with higher computational cost [4, 5, 6].

2.4 Indonesian NLP and Sentiment Analysis

Indonesian NLP research has improved through resources such as IndoNLU and IndoBERT [7], yet sentiment analysis still faces challenges from informal spelling, slang, and morphology. Previous Indonesian studies also report that BiLSTM is competitive, especially when preprocessing is carefully tailored to the language [8, 9, 10, 11].

2.5 App Review Sentiment Analysis

App-review sentiment analysis is an applied setting where deep models often perform well, but class imbalance remains a persistent issue [12, 13]. To the best of our knowledge, a direct benchmark between Scikit-learn classifiers and a PyTorch BiLSTM on large-scale Indonesian Spotify reviews with shared preprocessing has not been previously reported.

3 Dataset

3.1 Data Source and Collection

The dataset consists of 100,000 Indonesian-language Spotify reviews scraped from the Google Play Store using the google-play-scraper library. For modeling, only review text and star rating are used.

The raw dataset includes reviewer name, review text, rating, date, likes, and application version. Only the *Ulasan* and *Rating* columns are retained for sentiment classification.

3.2 Sentiment Label Construction

Following common rating-based labeling, ratings 1–2 are mapped to Negatif, rating 3 to Netral, and ratings 4–5 to Positif. This scheme matches the natural interpretation of app-store reviews.

3.3 Dataset Statistics and Class Distribution

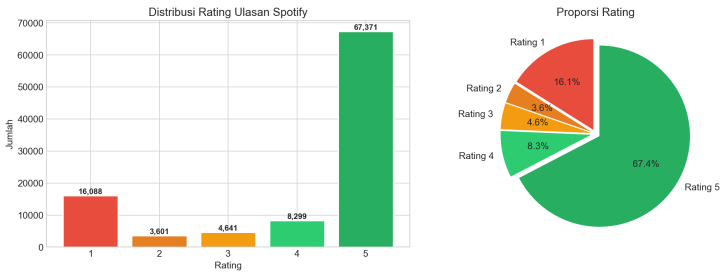

Figure 1: Sentiment distribution rating label from raw data.

The dataset is strongly imbalanced, with positive reviews dominating the corpus. After label mapping, deduplication, and removal of empty reviews, the cleaned dataset contains 70,155 samples: Positif 47,991 (68.4%), Negatif 18,026 (25.7%), and Netral 4,138 (5.9%).

Table 1: Dataset Statistics and Class Distribution

Split	Positive	Netral	Negative	Total
Full cleaned dataset	47,991 (68.4%)	4,138 (5.9%)	18,026 (25.7%)	70,155
ML training set	47,991	4,138	18,026	70,155
DL subset (20K stratified)	13,681	1,180	5,139	20,000
DL Train	9,852	851	3,700*	14,219*
DL Validation	∼1,739	∼150	∼653*	2,510*
DL Test	∼2,090	∼179	∼786*	2,953

Frequent negative themes include advertisements, premium access issues, and playback or lyric synchronization problems. These topics also appear in the cleaned vocabulary used by both tracks.

3.4 Preprocessing Pipeline

A unified preprocessing pipeline is applied to both tracks:

1. Lowercasing: all text is converted to lowercase.
2. Noise removal: URLs, mentions, hashtags, punctuation, symbols, and digits are removed.
3. Slang normalization: 124 informal forms are mapped to standard Indonesian (e.g., gak → tidak).
4. Stopword removal: 809 general and domain-specific stopwords are filtered out.
5. Morphological stemming: tokens are reduced to root forms using Sastrawi [11].

After preprocessing, the median review length is 4 tokens and the 95th percentile is 17 tokens. Therefore, the BiLSTM maximum sequence length is set to 17, and reviews that become empty after cleaning are removed.

Table 2: Preprocessing Pipeline Summary

Step	Method	Detail
Lowercasing	Python `str.lower()`	All characters converted to lowercase
Noise removal	Regex substitution	URLs, @mentions, punctuation, digits removed
Slang normalization	Custom dictionary	124 entries (e.g., *gak* → *tidak*)
Stopword removal	Sastrawi + custom	809 stopwords including domain terms
Morphological stemming	Sastrawi (ECS)	40+ suffix/prefix rules for Indonesian

4 Methodology

This study follows two parallel tracks — machine learning (ML) and deep learning (DL) — with the same preprocessing pipeline.

4.1 Machine Learning Track (Scikit-learn)

4.1.1 Feature Extraction: TF-IDF

Preprocessed reviews are converted into TF-IDF vectors with a maximum of 3,000 features [1].

4.1.2 Class Balancing: SMOTE

Because the dataset is imbalanced, SMOTE [13] is applied only to the training fold in each cross-validation split.

4.1.3 Classifiers

Three classifiers are compared: SVM, Multinomial Naive Bayes, and Decision Tree. All are implemented in Scikit-learn and evaluated with stratified 5-fold cross-validation on the cleaned dataset.

4.2 Deep Learning Track (BiLSTM)

4.2.1 Tokenization and Vocabulary

Texts in the DL track are tokenized at word level. The training split produces a 3,285-token vocabulary, and all sequences are padded or truncated to length 17.

4.2.2 Model Architecture

The BiLSTM model uses 128-dimensional embeddings, a two-layer bidirectional LSTM with 128 hidden units per direction, dropout, and a fully connected classification head. The total parameter count is 1,096,579.

4.2.3 Training Configuration

The model is trained on a stratified 20,000-sample subset with Adam optimizer, learning rate 10^{-3}, batch size 64, and unweighted cross-entropy loss. Early stopping selects the best checkpoint at epoch 3.

Using a 20,000-sample subset for the BiLSTM reflects CPU-only training constraints and is treated as a limitation in the comparison.

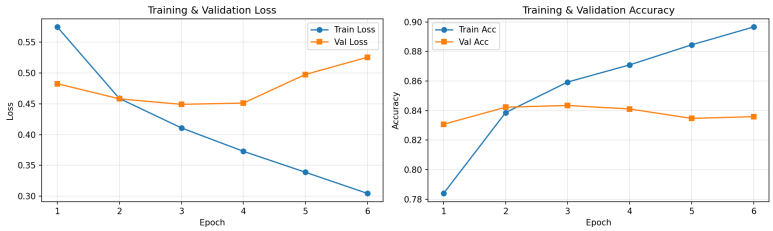

Figure 2: Training and Validation Loss/Accuracy Curves during the optimization phase.

5 Experiments

5.1 Experimental Setup

All experiments are conducted in a CPU-only environment (Intel-based local machine, PyTorch 2.2.2+cpu). The ML track runs on the full cleaned dataset of 70,155 samples using Scikit-learn 1.4.2 with PyCaret as the AutoML orchestration framework. The DL track trains the BiLSTM on a stratified 20,000-sample subset using PyTorch 2.2.2. All random states are fixed at seed=42 across both tracks for reproducibility.

5.2 ML Hyperparameter Configuration

Table 3 details the hyperparameter configuration for the three classical machine learning classifiers. All classifiers are evaluated via stratified 5-fold cross-validation on the full 70,155 training samples.

Table 3: ML Classifier Hyperparameter Configuration

Classifier	**Key Hyperparameters**	**Feature Representation**
Support Vector Machine	Kernel: linear; C: 1.0 (default)	TF-IDF, max features=3,000
Multinomial Naive Bayes	Alpha: 1.0 (Laplace smoothing)	TF-IDF, max features=3,000
Decision Tree	Criterion: gini; splitter: best	TF-IDF, max features=3,000

All classifiers are evaluated via stratified 5-fold cross-validation on the full 70,155 training samples. Performance metrics are macro-averaged across folds. Following benchmark evaluation, the best-performing model (Decision Tree) undergoes hyperparameter tuning via PyCaret's tune_model() with n_iter=50, optimizing for macro F1-Score, before finalization with finalize_model() (retraining on the full dataset).

5.3 DL Hyperparameter and Training Configuration

The BiLSTM is trained on a stratified 20,000-sample subset, split into training (14,219 / 72.1%), validation (2,510 / 12.7%), and test (2,953 / 15.0%) sets using stratified partitioning to preserve class proportions at each split. The vocabulary is built exclusively from the training split (minimum token frequency = 2), yielding a final vocabulary of 3,285 unique tokens.

Table 4 summarizes the training and architecture hyperparameters for the BiLSTM model.

Table 4: BiLSTM Training Hyperparameters

Hyperparameter	**Value**
Optimizer	Adam
Learning rate (η)	1×10^{-3}
Loss function	CrossEntropy Loss (unweighted)
Batch size	64
Max epochs	10
Early stopping patience	3 epochs (monitor: val_loss)
Max sequence length	17 tokens (95th percentile of training lengths)
Embedding dimension	128
LSTM hidden size	128 per direction (256 total)
LSTM layers	2 (bidirectional)
Dropout (post-LSTM)	0.5

Training terminates via early stopping at epoch 6 (best checkpoint restored from epoch 3, val_loss = 0.4491, val_acc = 0.8434). The training dynamics are visualized in Figure 3 training loss decreases monotonically from 0.5747 to 0.3046 across six epochs, while validation loss reaches its minimum at epoch 3 (0.4491) before diverging — a classic overfitting signature on a small training subset. Validation accuracy plateaus between 0.83–0.84 from epoch 2 onward, indicating the model saturates its representational capacity on 14,219 training samples.

5.4 Evaluation Metrics

Both tracks are evaluated using accuracy, weighted precision, weighted recall, weighted F1-score, Cohen's Kappa, and macro F1. Weighted F1 is the primary metric because the dataset is imbalanced, while macro F1 is used to inspect minority-class performance. The ML track additionally reports AUC from 5-fold cross-validation, and the DL track reports the same metrics on the held-out test set.

5.5 Fairness Considerations and Confounding Factors

The two tracks are not fully symmetric: ML models train on 70,155 samples, while the BiLSTM uses 14,219 training samples. This reflects a practical CPU-only setting, but all preprocessing steps remain identical across both tracks.

6 Results and Discussion

6.1 Machine Learning Results

Table 5 summarizes the 5-fold cross-validated performance of the three Scikit-learn classifiers on the full 70,155-sample corpus. Decision Tree achieves the highest scores across all primary metrics, attaining an accuracy of 0.7286, weighted F1-score of 0.7269, and Cohen's Kappa of 0.5928 — indicating substantial agreement beyond chance. SVM ranks second with accuracy 0.6991 and weighted F1-score 0.6958. Multinomial Naive Bayes performs markedly lower, achieving accuracy 0.4948 and weighted F1-score 0.4619. The complete benchmark is presented below.

Table 5: ML Cross-Validation Benchmark Results

Model	Accuracy	AUC	Recall	Precision	F1-Score	Kappa
Decision Tree	0.7286	0.8055	0.7286	0.7273	0.7269	0.5928
Support Vector Machine	0.6991	—	0.6991	0.6956	0.6958	0.5487
Multinomial Naive Bayes	0.4948	0.6899	0.4948	0.6181	0.4619	0.2422

All metrics are macro-weighted averages from 5-fold stratified cross-validation on 70,155 samples with SMOTE applied within each fold. AUC not available (—) for SVM under the PyCaret configuration used.

Decision Tree performs better than SVM on this dataset, likely because SMOTE and short review length favor simpler partitioning over sparse linear separation. Multinomial Naive Bayes performs worst, indicating that feature independence is too restrictive for colloquial Indonesian reviews.

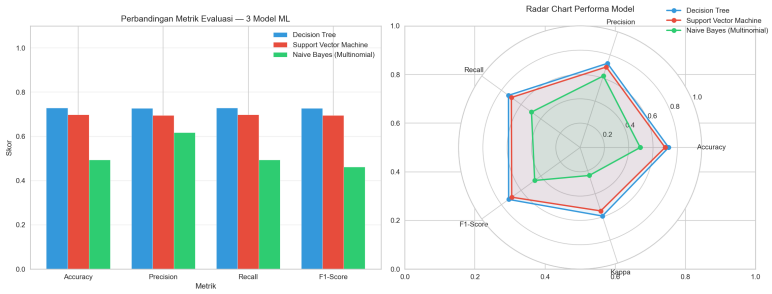

Figure 3: Performance comparison of the three Scikit-learn classifiers across four evaluation metrics.

The Decision Tree per-class report shows the best performance on Positif (F1 0.802), followed by Netral (0.717) and Negatif (0.707). This suggests that distinguishing Negative from Neutral remains the hardest case.

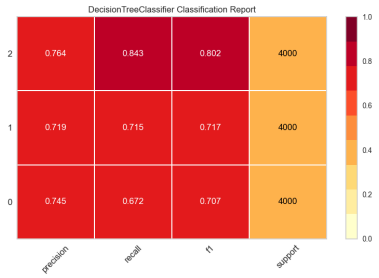

Figure 4: Per-class classification report of the Decision Tree classifier on the held-out test fold.

The confusion matrix confirms that most errors occur between Negative and Neutral, which are adjacent in the rating-based labeling scheme.

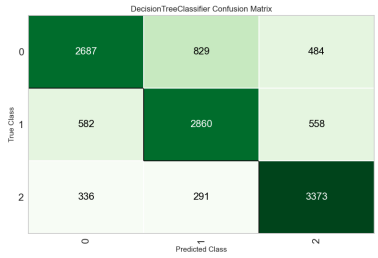

Figure 5: Confusion matrix of the Decision Tree classifier on the test fold (balanced via SMOTE).

6.2 BiLSTM Deep Learning Results

The BiLSTM model is evaluated on a held-out test set of 2,953 samples from the 20,000-sample subset. Table 6 reports the metrics at the best checkpoint (epoch 3).

Table 6: BiLSTM Test Set Evaluation Results

Metric	Value
Test Loss	0.4735
Accuracy	0.8314
Weighted Precision	0.7841
Weighted Recall	0.8314
Weighted F1-Score	0.8069
Macro F1-Score	0.5478
Total Parameters	1,096,579
Training Subset	14,219 samples

BiLSTM achieves weighted F1-score 0.8069, but its macro F1-score drops to 0.5478, indicating weak minority-class performance.

Table 7: BiLSTM Per-Class Classification Report

Class	Precision	Recall	F1-Score	Support
Negatif	0.7044	0.7807	0.7406	766
Netral	0.0000	0.0000	0.0000	176
Positif	0.8830	0.9234	0.9028	2,011
Weighted Avg	**0.7841**	**0.8314**	**0.8069**	**2,953**
Macro Avg	**0.5291**	**0.5680**	**0.5478**	**2,953**

The per-class report shows a clear imbalance: BiLSTM performs very well on Positif (F1 0.9028), reasonably on Negatif (F1 0.7406), and fails completely on Netral (F1 0.0000). The confusion matrix confirms that none of the 176 Neutral samples are classified correctly.

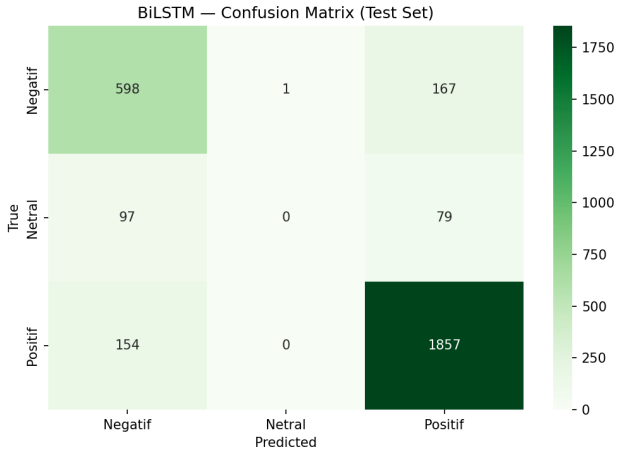

Figure 6: Confusion matrix of the BiLSTM model on the held-out test set ($n = 2,953$).

This Neutral-class collapse is mainly caused by class imbalance and the absence of class weighting or oversampling in the DL pipeline. The Neutral class is also semantically ambiguous, making it harder to separate from Negative and Positive in short informal reviews.

Figure 7 indicates early overfitting: training loss keeps decreasing, while validation loss reaches its minimum at epoch 3 and then rises. Validation accuracy stabilizes around 0.83–0.84, so early stopping selects epoch 3.

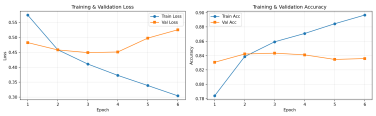

Figure 7: BiLSTM training and validation loss (left) and accuracy (right) across six epochs.

6.3 Comparative Analysis: ML vs. BiLSTM

Table 8: Head-to-head performance comparison of ML and BiLSTM models.

Model	Paradigm	Train Samples	Acc.	W-Prec.	W-Recall	W-F1	Macro-F1	Kappa	Params
Decision Tree	ML (sklearn)	70,155	0.7286	0.7273	0.7286	0.7269	∼0.72*	0.5928	—
SVM	ML (sklearn)	70,155	0.6991	0.6956	0.6991	0.6958	∼0.70*	0.5487	—
Mult. Naive Bayes	ML (sklearn)	70,155	0.4948	0.6181	0.4948	0.4619	∼0.46*	0.2422	—
BiLSTM	DL (PyTorch)	14,219	0.8314	0.7841	0.8314	0.8069	0.5478	—	1,096,579

Using weighted metrics, BiLSTM outperforms the best ML model. Its weighted F1-score is 0.8069 versus 0.7269 for Decision Tree, and its accuracy is 0.8314 versus 0.7286.

However, this advantage must be interpreted carefully because the two tracks are not symmetric. ML models are trained on 70,155 samples, whereas BiLSTM uses only 14,219 training samples. The comparison therefore reflects a practical compute-constrained setting rather than a perfectly controlled architectural test.

Macro F1 tells a different story. BiLSTM reaches only 0.5478 because it collapses on the Neutral class, while Decision Tree maintains more balanced per-class performance. In practice, BiLSTM is preferable when overall Positive/Negative discrimination is the main goal, whereas ML+SMOTE is more reliable for balanced three-class classification.

6.4 Discussion

Effect of class imbalance

Both tracks are affected by strong class imbalance. SMOTE helps the ML models retain Neutral-class performance, whereas the unweighted BiLSTM largely ignores that minority class.

Effect of training data asymmetry

The 4.9× training-data asymmetry prevents a fully equal comparison. Even so, BiLSTM still shows a strong advantage on weighted metrics, suggesting that sequential context is useful for short Indonesian reviews.

Preprocessing effectiveness

The shared preprocessing pipeline reduces vocabulary variation and keeps the final training vocabulary compact at 3,285 tokens. This benefits both TF-IDF features and embedding coverage.

Deployment and reproducibility

Both the Decision Tree and BiLSTM models are deployed on Hugging Face Spaces, enabling direct inspection of their behavior.

Limitations

The main limitations are training-data asymmetry, the absence of class balancing in the DL track, and different evaluation settings between ML cross-validation and DL hold-out testing. In addition, the study does not use pre-trained Indonesian embeddings.

7 Conclusion

This study compares three classical machine learning classifiers with a BiLSTM for three-class sentiment classification of Indonesian Spotify reviews. With 5-fold cross-validation on the full cleaned dataset, Decision Tree is the strongest ML model with weighted F1-score 0.7269. On its held-out test set, BiLSTM reaches weighted F1-score 0.8069 and accuracy 0.8314.

The main trade-off lies in class balance. BiLSTM gives better overall weighted performance, but it fails on the minority Neutral class. In contrast, Decision Tree with SMOTE produces more stable performance across all three classes. Therefore, BiLSTM is preferable for overall polarity detection, while ML+SMOTE is more suitable when balanced three-class recognition is required.

Future work should train the BiLSTM on the full corpus, add class balancing to the DL pipeline, and test pre-trained Indonesian embeddings or transformer-based models.

References

[1] Bo Pang and Lillian Lee. Opinion mining and sentiment analysis. *Foundations and Trends in Information Retrieval*, 2(1–2): 1–135, 2008. doi:10.1561/1500000011.

[2] Ananya Pratap and Nidhi. Comparative sentiment analysis of app reviews. *arXiv preprint arXiv:2006.09739*, 2020. URL `https://arxiv.org/abs/2006.09739`.

[3] Furqan Rustam, Imran Ashraf, Arif Mehmood, Saleem Ullah, et al. A comparative study of sentiment analysis using NLP and different machine learning techniques on US airline Twitter data. *arXiv preprint arXiv:2110.00859*, 2021. URL `https://arxiv.org/abs/2110.00859`.

[4] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi:10.1162/neco.1997.9.8.1735.

[5] Guixian Xu, Yuxin Meng, Xuxiang Qiu, Zhenlong Yu, and Xueliang Wu. Sentiment analysis of comment texts based on BiLSTM. *IEEE Access*, 7:51522–51532, 2019. doi:10.1109/ACCESS.2019.2911964.

[6] Chih-Hsueh Lin and Ulin Nuha. Sentiment analysis of Indonesian datasets based on a hybrid deep-learning strategy. *Journal of Big Data*, 10(1):88, 2023. doi:10.1186/s40537-023-00782-9.

[7] Bryan Wilie, Karissa Vincentio, Genta Indra Winata, Samuel Cahyawijaya, Xiaohong Li, Zhi Yuan Lim, Sidik Soleman, Rahmad Mahendra, Pascale Fung, Syafri Bahar, and Ayu Purwarianti. IndoNLU: Benchmark and resources for evaluating Indonesian natural language understanding. In *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing*, pages 843–857. Association for Computational Linguistics, 2020. URL `https://aclanthology.org/2020.aacl-main.85`.

[8] Setyo Kuncahyo et al. Improving Bi-LSTM performance for Indonesian sentiment analysis using paragraph vector. *arXiv preprint arXiv:2009.05720*, 2020. URL `https://arxiv.org/abs/2009.05720`.

[9] Muhammad Farizki et al. Klasifikasi sentimen menggunakan bidirectional LSTM dan IndoBERT dengan dataset terbatas. *ZONAsi: Jurnal Sistem Informasi*, 2025. URL `https://journal.unilak.ac.id/index.php/zn/article/view/25091`.

[10] Oluwaseun Abiola et al. Sentiment classification on the 2024 Indonesian presidential candidate dataset using deep learning approaches. *Indonesian Journal of Statistics and Its Applications*, 2024. URL `https://journal-stats.ipb.ac.id/index.php/ijsa/article/view/1259`.

[11] Mirna Adriani, Jelita Asian, Bobby Nazief, Saied M. M. Tahaghoghi, and Hugh E. Williams. Stemming Indonesian. *ACM Transactions on Asian Language Information Processing*, 6(4):1–33, 2007. doi:10.1145/1316457.1316458.

[12] Chukwuemeka Ivan Ossai and Nilmini Wickramasinghe. Sentiment analysis on Google Play Store app users' reviews based on deep learning approach. *Multimedia Tools and Applications*, 2024. doi:10.1007/s11042-024-19185-w.

[13] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi:10.1613/jair.953.